

RG – Uncertainty Geometry: The Forgetting Ladder

Where Mixing Locks, Why It Did Not, and the Operator Certificates That
Replace Sample Statistics

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Abstract

The previous uncertainty-geometry papers tried to lock *declaration forgetting* (mixing) with sample statistics on short ensembles, and it would not lock. This paper shows that the failure was structural and identifies the seven omitted steps. (1) Every invertible dynamics, including every finite Galerkin Navier–Stokes flow and the Sabra shell model, preserves total variation between propagated laws *exactly*; full-state TV forgetting is impossible there. (2) Mixing is pointwise in the law: over all absolutely continuous declarations the propagated TV diameter equals 1 at every finite time, so no uniform frontier exists without a *declared regularity class*. (3) TV-equivalence cannot be certified from finite samples at all without such a class; locks must be stated on operators, and samples serve only to refute. (4) The requested functional (fixed-time law versus time average) decides which ergodic property is needed. (5) Forgetting has four distinct mechanisms—expansion on a regularity class, migration into observation fibres, dissipative contraction, and noise—each with its own certificate. (6) Floating-point declarations are atomic and finite invertible computations recur. (7) A vanishing declaration diameter does not certify accuracy. With these steps restored, mixing locks: for the tent map and, by conjugacy, the logistic map at $r = 4$, $\text{TV}(T_{\#}^n \mu_f, T_{\#}^n \mu_g) \leq 2^{-(n+2)} \text{Var}(f - g)$, with the constant *attained* at every n . For two-dimensional Galerkin Navier–Stokes, forgetting locks in Wasserstein distance below an explicit Galerkin–Grashof gate $G_N = S\|f\|/\nu^2 < 1$; the prior forced regime has $G_N \approx 2.8 \times 10^3$ and lies outside every route on the ladder, which is now a theorem rather than an unexplained null result. The rungs are placed as Stratum 7 (R30–R47) of the Theorem Ladder. A hostile battery of fifteen attack modules, four seeds, exact rational and integer arithmetic where the operator allows it, and a mandatory mutation for every test was run; all fourteen ladder targets survived with demonstrated sensitivity, and five prior claims or audits were broken.

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1 The diagnosis: seven omitted steps

The prior chain was: *chaos* $\Rightarrow$ *ensembles forget their declaration* $\Rightarrow$ *the forecast target is forced* $\Rightarrow$ *Monte Carlo is objective there*. Each arrow hides a step.

Omitted step 1 (Invertibility). For a bimeasurable bijection Φ , $\text{TV}(\Phi_{\#}\mu, \Phi_{\#}\nu) = \text{TV}(\mu, \nu)$ (R33). A smooth ODE flow is a bijection onto its image. Hence the Galerkin NS and Sabra runs could never exhibit full-state TV forgetting, whatever the Reynolds number, horizon, or ensemble size. Forgetting in those systems can only occur on an observation quotient (R34) or in a weak metric (R39).

Omitted step 2 (Uniformity). “Forced over an admissible family” is a supremum statement. Mixing gives $\text{TV}(P^n\mu, P^n\nu) \rightarrow 0$ for each pair, but the supremum over all absolutely continuous laws is 1 for every n (R36). The regularity class that makes a frontier exist was never declared; the finite list of four laws silently stood in for it.

Omitted step 3 (Certificate locus). TV-equivalence cannot be certified by any finite-sample procedure without a regularity class (R43, Le Cam). W1 can, but needs $m \geq 2162$ members in the prior design. A lock must be an inequality on the transfer operator, flow, or kernel. Samples are a refutation channel.

Omitted step 4 (Functional type). For a time-average functional, unique ergodicity forgets *every* declaration, including Dirac masses, at an explicit rate; for a fixed-time law, mixing on a class is required (R41). The prior “periodic involution retains the declaration” is true only for the fixed-time reading.

Omitted step 5 (Mechanism). Four mechanisms forget, with four different certificates: expansion and folding on a regularity class (R37–R38), migration of the declaration difference into observation fibres (R34, R42), dissipative contraction (R39), and noise (R40). The prior PDE runs had negative separation slopes, which signals the contraction mechanism; the contraction certificate was never computed, and it fails at those parameters.

Omitted step 6 (Finite state). A float is a dyadic rational. The float tent map is exact arithmetic, not rounding, and every dyadic rational is eventually fixed; the sampled law is atomic and outside every BV class (R36, A07). Finite invertible computations are permutations and recur exactly (R35).

Omitted step 7 (Accuracy). Two ensembles can agree perfectly while both are wrong: in A07 the declaration diameter is 0 and both empirical laws are at TV distance 0.98 from the true pushforward (R47).

2 Declarations

Rung R30 (Experiment declaration — declared). *An uncertainty experiment is*

$$E = (X, \Phi_t, O, \mathsf{T}, \mathcal{A}, d, \tau) :$$

state space, dynamics, observation, functional type T (fixed-time law, time average, ...), admissible declaration class $\mathcal{A}$, metric d , task tolerance τ .

Rung R31 (Declaration diameter — forced given E). $\mathcal{D}_t(\mathcal{A}) = \sup_{\mu, \nu \in \mathcal{A}} d((O \circ \Phi_t)_\# \mu, (O \circ \Phi_t)_\# \nu)$ for fixed-time laws, and the analogous supremum of $|\langle \cdot \rangle_t^\mu - \langle \cdot \rangle_t^\nu|$ for time averages. The output is forced at scale τ from time t^* if $\mathcal{D}_t(\mathcal{A}) \leq \tau$ for all $t \geq t^*$. The supremum over $\mathcal{A}$, not a finite sample of $\mathcal{A}$, is the object.

Rung R32 (Observation monotonicity — imported). For any Markov kernel K (in particular any measurable map), $\text{TV}(\mu K, \nu K) \leq \text{TV}(\mu, \nu)$; for O L -Lipschitz, $W_1(O_\# \mu, O_\# \nu) \leq L W_1(\mu, \nu)$. Hence $\mathcal{D}_t^O \leq \mathcal{D}_t^{\text{id}}$.

3 The no-go stratum

Rung R33 (TV rigidity — no-go). If Φ is a bimeasurable bijection, $\text{TV}(\Phi_\# \mu, \Phi_\# \nu) = \text{TV}(\mu, \nu)$ for all laws.

Proof. $\text{TV}(\Phi_\# \mu, \Phi_\# \nu) = \sup_B |\mu(\Phi^{-1}B) - \nu(\Phi^{-1}B)|$ and $B \mapsto \Phi^{-1}B$ is a bijection of the σ -algebra. $\square$

In RG language: the kernel pair of a bijection is the diagonal, so nothing becomes unresolved; R33 is the measure-level shadow of a trivial R_Φ (R1) and the dynamical analogue of the entropy no-go R26. *Attack A01* (exact integer pushforwards on the 64×64 cat lattice, period 48, four seeds, all n up to the period) found no violation; a non-injective mutation is detected at $n = 1$. *Attack A02* confirms numerically that the forced 2D Galerkin NS at the prior parameters ($\nu = 0.025$, 0.05 Taylor–Green forcing, $T = 2$) is injective: forward–backward round trips recover the state to 1.2×10^{-7} , 3.9×10^{-9} , 1.2×10^{-10} at $\Delta t = 0.04, 0.02, 0.01$ (halving ratio ≈ 32 : convergent), while a non-injective mutation leaves error 0.21.

Rung R34 (Forgetting is fibre migration — forced). $(O \circ \Phi_t)_\# \mu = O_\#(\Phi_t)_\# \mu$ and $R_{O \circ \Phi_t} = (\Phi_t \times \Phi_t)^{-1} R_O$. For bijective Φ_t the fibres of the pulled-back observation are the Φ_t^{-1} -images of the fibres of O : observed forgetting at time t is exactly the declaration difference lying inside the fibres of $O \circ \Phi_t$ at time 0. In particular $t \mapsto \mathcal{D}_t^O$ need not be monotone.

Attack A03 checks the identity in exact integers for all n (block observation 8×8); the observed TV reaches exactly 0 at $n = 6$ and returns to 1 (Fig. 2). *Attack A08* gives an exact unforgetting witness on the continuum cat map: a declaration difference in Fourier mode $k = A^m k_0$ is invisible in the window $|k|_\infty \leq 2$ at $t = 0$ and visible at $t = m$ (e.g. $k = (39603, 24476)$, $m = 11$). An initially matched observable can acquire declaration dependence; the NS paper treated this as an empirical question, but it is structural.

Rung R35 (Finite-state recurrence and collapse — no-go). A deterministic map on a finite set is eventually periodic; a bijection is periodic with some period p , so $\mathcal{D}_{n+p}^O = \mathcal{D}_n^O$ exactly. A non-bijective one maps every law into the cycle set. No finite-state deterministic computation certifies permanent forgetting.

Rung R36 (No uniform forgetting without declared regularity — no-go). Let $O \circ \Phi_t$ be continuous and non-constant and let $\mathcal{A}$ contain the uniform laws on all small balls. Then $\mathcal{D}_t^{\text{TV}}(\mathcal{A}) = 1$ for every t .

Proof. Take x_1, x_2 with distinct images $y_1 \neq y_2$, disjoint neighbourhoods $V_i \ni y_i$, and open $U_i \ni x_i$ mapped into V_i . Uniform laws on U_1 and U_2 have pushforwards with disjoint supports. $\square$

For the exact (hence mixing) tent map the witness is explicit: $f_k = 2^k \mathbf{1}_{[0, 2^{-k}]}$ has $P^n f_k = 2^{k-n} \mathbf{1}_{[0, 2^{n-k}]}$ and $\text{TV}(P^n f_k, 1) = 1 - 2^{n-k}$ for $n \leq k$. *Attack A04* verifies this closed form in exact rationals for $k \leq 14$ (Fig. 1, right). This is the theorem behind “mixing won’t lock”: no amount of chaos replaces a declared class.

4 The lock stratum

Rung R37 (Compactness lock — gated: exactness, compact declaration). *If the transfer operator P is exact ($\|P^n f - h\|_1 \rightarrow 0$ for every density f) and $\mathcal{A}$ is relatively compact in L^1 , then $\mathcal{D}_n^{\text{TV}}(\mathcal{A}) \rightarrow 0$.*

Proof. P is an L^1 contraction. Given ϵ , take a finite ϵ -net $\{f_i\}$; then $\|P^n f - h\|_1 \leq \epsilon + \max_i \|P^n f_i - h\|_1$. $\square$

Compactness gives the lock; it does not give the rate. The rate requires a quantitative class.

Rung R38 (BV mixing lock, sharp — gated: BV declaration). *For the tent map $T(x) = 1 - |2x - 1|$ and densities $f, g \in BV[0, 1]$,*

$$\text{TV}(T_{\#}^n \mu_f, T_{\#}^n \mu_g) \leq 2^{-(n+2)} \text{Var}(f - g) \quad (n \geq 0),$$

and equality holds for every n at $f = 2^{n+1} \mathbf{1}_{[0, 2^{-n-1}]}$, $g = 1$.

Proof. $Pf(x) = \frac{1}{2}[f(x/2) + f(1 - x/2)]$. (L1) The two branches reparametrize f on $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ monotonically, so $\text{Var}(Pf) \leq \frac{1}{2}(\text{Var}_{[0, 1/2]} f + \text{Var}_{[1/2, 1]} f) = \frac{1}{2} \text{Var} f$. (L2) For zero-mean h with $M = \sup h \geq 0 \geq m = \inf h$ and $p = |\{h > 0\}|$, $\|h\|_1 = 2 \int h_+ = 2 \int h_- \leq 2 \min(Mp, |m|(1-p)) \leq 2M|m|/(M + |m|) \leq (M + |m|)/2 \leq \text{Var}(h)/2$. (L3) P preserves the mean and $\text{TV} = \frac{1}{2} \|\cdot\|_1$. Hence $\text{TV}_n \leq \frac{1}{4} \text{Var}(P^n(f - g)) \leq 2^{-(n+2)} \text{Var}(f - g)$. For the witness, $\text{Var}(f - 1) = 2^{n+1}$ and $P^n f = 2 \cdot \mathbf{1}_{[0, 1/2]}$, whose TV from 1 is $\frac{1}{2} = 2^{-(n+2)} 2^{n+1}$. $\square$

The naive estimate $\|h\|_1 \leq \text{Var} h$ gives $2^{-(n+1)}$; the equality witness shows that this is off by exactly a factor 2, and (L2) closes the gap, so no smaller constant is possible. Consequences:

Corollary 1 (Exact uniform frontier). *On $\mathcal{A}_V = \{\text{densities} : \sup \text{Var}(f - g) \leq V\}$, $\mathcal{D}_n \leq 2^{-(n+2)} V$, so the output is forced at scale τ from $n^* = \lceil \log_2(V/\tau) \rceil - 2$. For $V = 2^k$ and $\tau = \frac{1}{2}$ this frontier is exact: the witness has $\text{TV} \frac{3}{4}$ at $n^* - 1$.*

Corollary 2 (Logistic map, by conjugacy). *$h(x) = \sin^2(\pi x/2)$ satisfies $L \circ h = h \circ T$ for $L(y) = 4y(1 - y)$. For laws ρ, σ on $[0, 1]$ with pulled-back densities $f_\rho = (\rho \circ h)h'$, $\text{TV}(L_{\#}^n \rho, L_{\#}^n \sigma) \leq 2^{-(n+2)} \text{Var}(f_\rho - f_\sigma)$, because TV is invariant under the bijection h (R33 used positively).*

Attack A05 (exact rationals, dyadic level 10; random, spiky, sparse, monotone and oscillating declarations; four seeds; adversarial hill-climbing re-verified in exact arithmetic) found no violation; the largest adversarial ratio was 0.336, the equality witness reaches 1, and the overclaim $2^{-(n+3)}$ is refuted by the battery's own witness. Each link (L1)–(L3) is tested separately at every step. *Attack A06* verifies the conjugacy to 8×10^{-16} and evaluates prior-style logistic declarations (uniform, Gaussian, bimodal; parameters declared here because the prior script was not supplied): $\max \text{Var}(f_\rho - f_\sigma) = 20.32$, so the certified population TV after 60 steps is below 4.5×10^{-18} .

Rung R39 (Galerkin NS contraction lock — gated: energy ball, $G_N < 1$). *For the exact-dealiased 2D Galerkin NS on the mean-zero torus, with support $\mathcal{K}$, $S = (\sum_{k \in \mathcal{K}} |k|^2)^{1/2}$ and $\|u\|^2 = \sum_k |\hat{u}_k|^2$: every solution with $\|u_0\| \leq R_0$ satisfies $\|u(t)\| \leq \rho(t) = e^{-\nu t} R_0 + \frac{\|f\|}{\nu} (1 - e^{-\nu t})$, and*

$$W_1(\Phi_{t\#} \mu, \Phi_{t\#} \nu) \leq \Lambda(t) W_1(\mu, \nu), \quad \Lambda(t) = \exp \left(-\nu t + S \int_0^t \rho(s) ds \right),$$

for every pair of declarations supported in the ball. Forgetting is eventually certified iff $G_N := S\|f\|/\nu^2 < 1$; for $f = 0$ it is always eventually certified.

Proof. Energy: $\frac{d}{dt} \frac{1}{2} \|u\|^2 = -\nu \|\nabla u\|^2 + \langle f, u \rangle \leq -\nu \|u\|^2 + \|f\| \|u\|$ (Poincaré, $|k| \geq 1$); the ball is forward invariant and convex. Linearization $\dot{w} = P_N[-(u \cdot \nabla)w - (w \cdot \nabla)u] + \nu \Delta w$: exact dealiasing makes $\langle (u \cdot \nabla)w, w \rangle = 0$, and $|\langle (w \cdot \nabla)u, w \rangle| \leq \|\nabla u\|_\infty \|w\|^2$ with $\|\nabla u\|_\infty \leq S \|u\|$ by Cauchy–Schwarz on the finite support. So $\frac{d}{dt} \|w\|^2 \leq 2(-\nu + S\rho) \|w\|^2$ along every trajectory in the ball. The segment between two initial states lies in the ball, so $\text{Lip}(\Phi_t) \leq \Lambda(t)$, and coupling transports the bound to W_1 . $\square$

TV stays constant meanwhile (R33): contraction moves mass closer, it does not merge it. *Attack A09*: 16^2 grid, $\nu = 0.1$, unforced, 4 families $\times 6$ members, four seeds, $T = 120$; every member pair at every saved time satisfies the bound (largest ratio 0.0072); the certified family diameter is $W_1 \leq 1.4 \times 10^{-6}$ at $T = 120$, frontier $t^* = 80$ for $\tau = 10^{-4}$, uniformly over all declarations on the perturbation sphere, with no samples. A sharpness pair ($R_0 \rightarrow 0$, $|k| = 1$ mode) attains 0.99953 of $\Lambda(t)$; the overclaim $e^{-2\nu t}$ is refuted. At the prior parameters, $S = 49.19$, $\|f\| = 0.0354$, $\nu = 0.025$ give $G_N = 2.78 \times 10^3$ and a certificate exponent rate at least +35.6, while the measured finite-window slope is -0.31 : a negative slope is not a contraction certificate.

Rung R40 (Markov lock — gated: noise; imported Dobrushin).

$$\text{TV}(\mu K^n, \nu K^n) \leq \delta(K^m)^{\lfloor n/m \rfloor} \text{TV}(\mu, \nu)$$

with $\delta(K) = \frac{1}{2} \max_{i,j} \|K(i, \cdot) - K(j, \cdot)\|_1$, attained at $n = m$ by Dirac pairs. Every deterministic kernel with two distinct images has $\delta = 1$.

Attack A10 broke the first formulation: for a rotation followed by local noise, the *one-step* coefficient is exactly 1 (disjoint row supports), so “noise gives $\delta < 1$ ” is false. The rung now uses blocks: first contraction at $m = 5$, best declared block $m = 60$ with $\delta(K^{60}) = 0.339$. A non-chaotic rotation forgets once noise is present; the chaotic invertible cat map never does in TV.

Rung R41 (Functional type — forced typing; imported Weyl). *Time averages*: for $x \mapsto 1 - x$, $|\langle x \rangle_n - \frac{1}{2}| \leq 1/(2n)$ (equality at $x = 0$, n odd); for rotation by α , $|\langle \cos 2\pi x \rangle_n| \leq 1/(n|\sin \pi \alpha|)$; both uniform over all declarations including Diracs. *Fixed-time laws of the same systems never forget*. *Chaos is not uniquely ergodic*: the logistic Dirac at $\frac{3}{4}$ has time average $\frac{3}{4}$.

Attack A11 confirms all bounds over 5000 initial points per seed up to $n = 10^4$.

Rung R42 (Weak forgetting of invertible hyperbolic maps — gated: band limit). *For the cat map A and densities band-limited to $|k|_\infty \leq K_0$, the observed Fourier window $|k|_\infty \leq K$ reads initial mode $A^n k$ at time n ; the declaration is forgotten exactly on the window from a finite $n^*(K_0, K)$ growing like $\log K / \log \lambda$, $\lambda = (3 + \sqrt{5})/2$, while TV is constant.*

Attack A08: exact integers give $n^* = 3, 4, 5, 7, 8, 10$ for $K = 1, 4, 16, 64, 256, 1024$ (Fig. 2, right). The frontier is a property of the observation scale.

5 The certification and placement stratum

Rung R43 (Samples refute, operators lock — imported + no-go). (a) *For bounded scalar outputs the simultaneous DKW certificate brackets W_1 at level $1 - \alpha$ (imported).* (b) *No finite-sample procedure certifies $\text{TV} < \frac{1}{2}$ without a regularity class: with P uniform on M cells and Q_S uniform on a random half, $\text{TV}(P, Q_S) = \frac{1}{2}$ yet any test from m draws has power at most $\alpha + 1.5 m^2/M$.*

Proof of (b). Conditioned on no repeated cell, both $P^{\otimes m}$ and the S -mixture of $Q_S^{\otimes m}$ are uniform over ordered distinct m -tuples; collision probabilities are at most $m^2/(2M)$ and m^2/M . So some S makes the test nearly blind; M is arbitrary. $\square$

Attack A12 measures simultaneous coverage 0.969–0.974 at nominal 0.95 (dropping the union bound gives 0.869, detected), reproduces the prior constants $\epsilon_m = 0.670984$ and $m \geq 2162$, and observes test power 0.027–0.060 against $\text{TV} = \frac{1}{2}$ for $M = 2^{40}$, $m = 500$, while the same test at $M = 64$ has power above 0.9.

Rung R44 (Monte Carlo placement — imported). *Pointwise $N^{-1/2}$ convergence for each declaration says nothing about $\mathcal{D}(\mathcal{A})$ (prior theorem, retained). Monte Carlo is the integration engine for a target whose invariance has been locked by R37–R42; it is not the lock.*

Rung R45 (Wall correspondence — forced). *At a fold $y = b^2 + \epsilon$, the full preimage of a y -confidence set covers b iff the set covers b^2 ; its coverage equals the observation coverage identically (R2–R3).*

Attack A13: identity in every one of 20000 replications per case, including the wall $b = 0$.

Rung R46 (Forced deterministic turbulence — open). *For forced deterministic NS above the gate, no route of this ladder applies: TV is rigid (R33), R39 requires $G_N < 1$, R40 requires noise, and R37–R38 require a non-invertible expanding transfer operator. With stochastic forcing, ergodicity of 2D NS is established qualitatively [7]; an explicit uniform constant for a declared family, which is what a lock needs, is open here.*

Rung R47 (Accuracy gate — gated / no-go). *$\mathcal{D}(\mathcal{A}) \leq \tau$ does not certify the target. A lock additionally needs an operator-approximation bound on the declared class, e.g. $\|P_{\text{num}}^n - P^n\|_{\mathcal{A} \rightarrow L^1}$.*

Attack A07: float and exact rational tent orbits coincide for 70 steps on 200 test points; all 2×10^5 samples per ensemble reach exactly 0 by $n = 63$ –66; the empirical diameter is 0 and the TV to the true pushforward is 0.984.

6 Placement on the Theorem Ladder (Stratum 7)

Rung	Statement	Type	Depends	Attack
R30	Experiment $E = (X, \Phi, O, \mathbb{T}, \mathcal{A}, d, \tau)$	declared	R0	—
R31	Declaration diameter $\mathcal{D}_t(\mathcal{A})$ as a supremum	forced (given R30)	R30	—
R32	Observation monotonicity (TV, Lipschitz W_1)	imported	R31	A01
R33	TV rigidity under bijections; ODE flows (NS, Sabra)	no-go	R1, R26	A01, A02
R34	Forgetting = fibre migration; non-monotone	forced	R1–R3	A03, A08
R35	Finite-state recurrence / collapse	no-go	R33	A03, A07
R36	No uniform forgetting without declared regularity	no-go	R31	A04, A07
R37	Exactness + L^1 -compact class $\Rightarrow$ uniform forgetting	gated	R36	A04, A05
R38	BV lock $2^{-(n+2)}\text{Var}$, sharp; logistic by conjugacy	gated (lock)	R37, R33	A05, A06
R39	Galerkin NS W_1 contraction lock, gate $G_N < 1$	gated (lock)	R32	A09

Rung	Statement	Type	Depends	Attack
R40	Dobrushin block lock; deterministic $\delta = 1$	gated / imported	R32	A10
R41	Functional type: time averages vs fixed-time laws	forced typing	R30	A11
R42	Cat-map weak forgetting, scale-bound frontier	gated	R34	A08
R43	Samples refute; TV not sample-certifiable	imported + no-go	R36	A12
R44	Monte Carlo is the engine, not the lock	imported	R43	A11,A12
R45	Wall correspondence = fibre saturation	forced	R2,R3	A13
R46	Forced deterministic turbulent forgetting	open	R33, R39, R40	A09 (gate)
R47	Accuracy gate: small $\mathcal{D}$ does not certify the target	gated / no-go	R31	A07

7 The hostile battery

Rules. Every module states a target, attempts to break it, and must also run a *mutation* that a working test is required to detect. A rung is accepted only if it survives and its mutation is detected; a test that cannot fail does not count. Seeds are 20261010, 1, 42, 777. Transfer operators of the tent and cat maps are evaluated in exact rationals and integers; floating point is used only where labelled, and numerical gates are tied to the margin of the claim, not to fixed tolerances.

The battery failed itself once. The first complete run reported A09 as broken: its timestep gate used an arbitrary tolerance 10^{-6} on the full state, and the halving error was 1.5×10^{-6} . The gate was replaced by the requirement that the integration error on the tested quantity (pair distances) be smaller than the claim’s margin; the margin was 0.993 for the ensemble and 4.7×10^{-5} for the sharpness pair, against errors 7.8×10^{-7} and 1.5×10^{-11} . A10 also broke its first formulation (one-step Dobrushin coefficient); the rung was corrected, not the test.

Module	Rung	Outcome	Mutation detected
A01 TV rigidity (cat lattice)	R33	survived	yes
A02 NS Galerkin injectivity	R33	survived	yes
A03 fibre migration, recurrence	R34, R35	survived	yes
A04 no uniform forgetting	R36	survived	yes
A05 BV lock, sharp	R38	survived	yes
A06 logistic transport	R38	survived	yes
A07 atomic floats, accuracy	R36, R47	survived	yes
A08 cat weak forgetting	R42, R34	survived	yes
A09 NS contraction lock and gate	R39	survived	yes
A10 Dobrushin block lock	R40	survived	yes
A11 functional type	R41	survived	yes
A12a/b sampling certificate, Le Cam	R43	survived	yes
A13 wall correspondence	R45	survived	yes

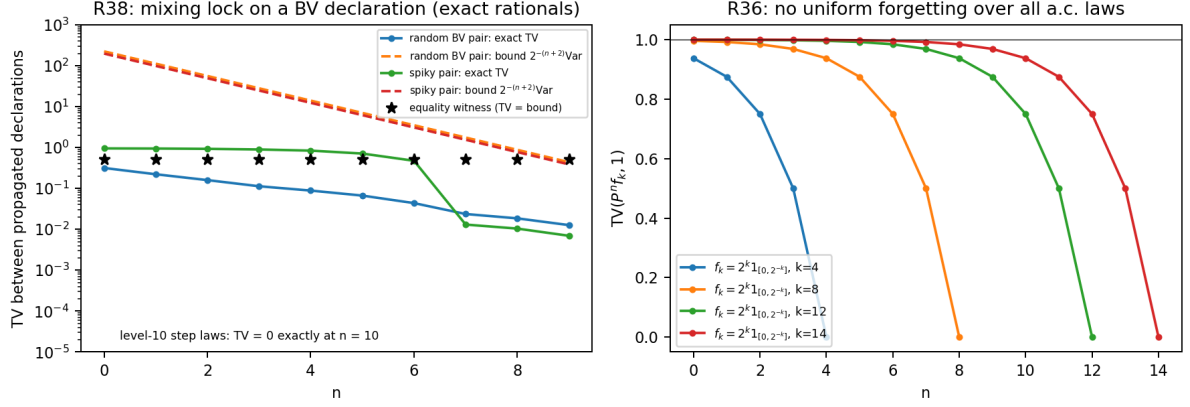


Figure 1: Left: R38 in exact rationals; the equality witness meets the bound at every n . Right: R36; the concentrating family keeps TV near 1 for any fixed n .

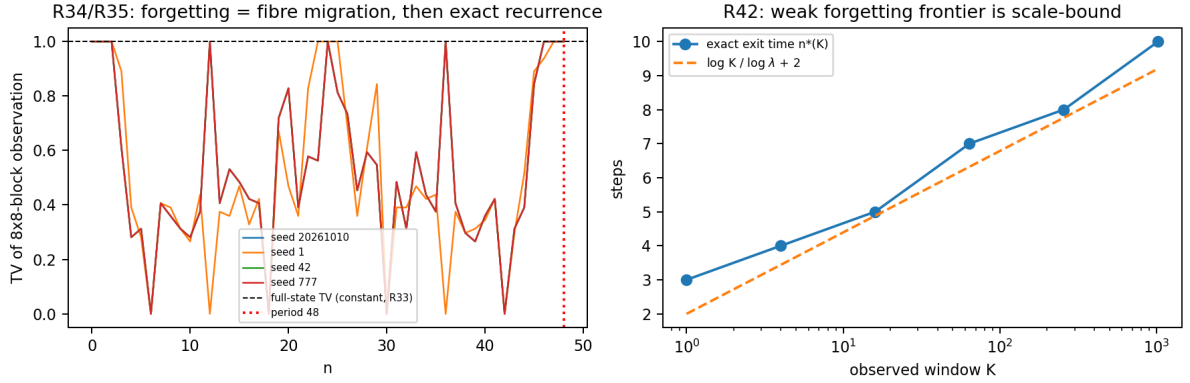


Figure 2: Left: R34/R35 on the cat lattice; full-state TV is constant, observed TV falls to exactly 0 and recurs at the period 48. Right: R42 exit times grow like $\log K / \log \lambda$.

Prior claims broken. **A06b** (Stress paper, Prop. 1): the reported late-window TV 0.02143, W_1 0.00442 and entropy spread 0.00861 cannot be population distances; the population TV is certified below 4.5×10^{-18} . **A12c**: the reported growth from 0.00877 to 0.04160 between 8 and 512 bins lies on the same-law null of the plug-in TV for about 5×10^4 – 10^5 samples per ensemble (Fig. 3); it is the estimator’s noise floor, not a scale effect. The prior *conclusion* (logistic forgets, involution does not) is true, and is now forced by R38 and R33. **A11b**: under a time-average reading the involution’s long-run mean is $\frac{1}{2}$ for every declaration; the reported 0.2 versus 0.8 holds only for a fixed-time functional, which was undeclared. **A14**: of the nine canaries in the supplied `ns_uncertainty_battery.py`, four are tautologies (`len(FAMILIES[:1]) < 4`, `len([2]) < 3`, a literal rank, and `-ns.nu < 0`), one rereads stored output, one tests literal fixture arrays, and three test the pipeline: 12 of the 36 reported trials could fail. **A15**: the supplied ZIP does not reproduce `RG_NS_Audited_Ladder.pdf`; scripts, member count (8 vs 12), ν (0.04 vs 0.025), horizon (1 vs 2), timestep, family names, 3D grids and Sabra viscosity all differ. **Retained**: the DKW arithmetic, the fibre-factorization theorem, and the Monte Carlo rate theorem survived (A12a, R44).

8 Mature operational rule

1. Declare E , including the functional type and the admissible *class*, not a list.

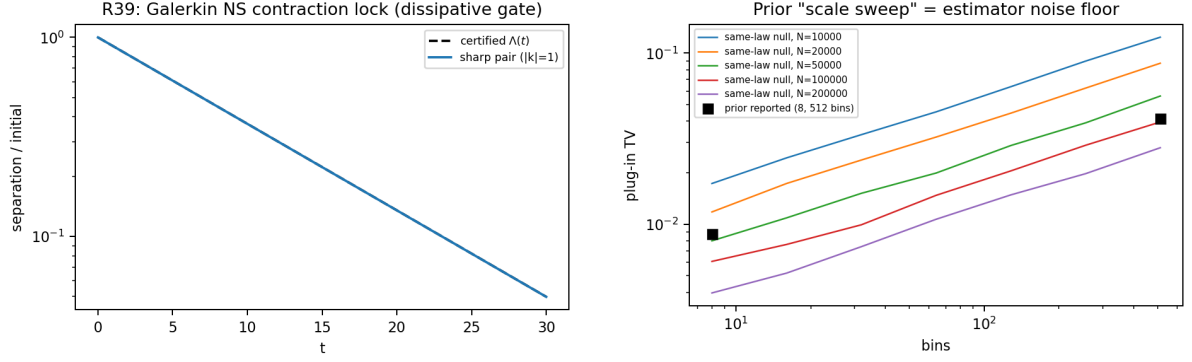


Figure 3: Left: R39; the sharpness pair is indistinguishable from the certified $\Lambda(t)$ (ratio ≥ 0.9995). Right: prior scale-sweep values against the same-law null of the plug-in TV.

2. Identify the mechanism: invertible (TV rigid; use observation fibres or a weak metric), expanding non-invertible (BV/cone lock), dissipative (contraction lock, check G_N), noisy (Dobrushin block lock).
3. Prove the operator inequality and its frontier on the class. That is the lock.
4. Attack it: exact arithmetic where possible, adversarial search, equality witnesses, and a mutation the test must detect.
5. Certify accuracy of the numerical operator separately (R47).
6. Use Monte Carlo only to integrate the forced target, and samples only to refute.

9 Non-claims

Non-claim 1. No lock is issued for forced deterministic Navier–Stokes at moderate or high Reynolds number, or for the prior PDE parameters; R46 is open, and R33 forbids full-state TV forgetting there for any parameters.

Non-claim 2. R39 is a crude, rigorous gate; it is sharp only as $R_0 \rightarrow 0$ and is loose by six orders for the declared high-band perturbations. Sharper 2D estimates are not attempted.

Non-claim 3. The logistic prior-style declarations in A06 use parameters chosen here; the prior script was not supplied. The bound is uniform over all BV pull-backs, so the conclusion does not depend on that choice.

Non-claim 4. The numerical attacks test the theorems; they do not replace the proofs, and A02 illustrates, rather than proves, injectivity of the exact flow, which follows from ODE uniqueness.

10 Conclusion

Mixing would not lock because it was asked to do something it cannot do: forget uniformly over an undeclared class, in a metric that invertible dynamics preserve, certified by samples that cannot certify it. Restoring the omitted steps gives a ladder on which mixing does lock, with a sharp constant, for the systems where it is true; contraction locks the dissipative Navier–Stokes regime; and the forced turbulent regime is marked open by theorem rather than by a failed experiment.

References

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A Reproducibility

```
python run_all.py --out results      # 15 modules, ~100 s on one core
python make_figures.py
```

Outputs: `results/ug_results.json` with every witness, seed, mutation result, environment and source SHA-256. The folder `prior/` contains the supplied files that A14 and A15 audit.